A POSSIBLE SOURCE IN CAMBODIA FOR AUSTRALASIAN TEKTITES

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A possible crater representing the source of Australasian tektites is identified in northeastern Cambodia at longitude, 106°34'E., and latitude, 13°55'N. The crater is an incomplete oval ring of hills 10 km long and 6 km wide located near the center of the Muong Nong-type tektite strewnfield. The morphology of the structure may have been significantly changed by relatively recent erosion and deposition processes. The types of rocks in the area of the crater are consistent with the compositional requirements for the tektite source rocks. Collection of impactite material from the vicinity of the structure will be required to confirm its association with Australasian tektites.

INTRODUCTION

The source of Australasian tektites has not been identified – yet. It is generally agreed that tektites are glasses solidified from rock melts produced during impact events large enough to form craters more than 10 km in diameter (King, 1977). A small minority believes tektites may be produced by volcanic processes (O'Keefe, 1976). The location of the Australasian tektite-forming impact or impacts has been a controversial question for some time. Sites in the South China Sea, in southeast Asia, under the Antarctic ice, and on the Moon have been suggested. The most recent proposals for the source of these tektites are the Elgygytgyn crater in northeast Siberia (Dietz, 1977) and the Zhamanshin crater located about 200 km north of the Aral Sea (Glass, 1978).

It is the purpose of this note to identify another, more acceptable, site as a source of Australasian tektites. The site we propose is located in northeast Cambodia at 106°34'E. longitude and 13°55'N. latitude.

DESCRIPTION OF AIRBORNE OBSERVER

While serving in the United States Air Force in southeast Asia in 1972 and 1973, one of us (ARR) routinely flew missions across northeastern

Cambodia as the pilot of an F-4 aircraft. From the flight altitude of 15,000 to 20,000 feet (4.6 to 6.1 km) the feature appears as a ring, about 5 to 10 km in diameter, of steep-walled hills rising out of a wide flat expanse with little or no topographic relief, other than the very obvious ring. Indeed, it was its amazing degree of apparent circular symmetry which attracted one's attention. Ten to 20 km to the east is a rugged mountain chain, much of which is known to be karst. The ring is not obviously associated with the mountains. To the southwest over 150 km is a 100-km-long, shallow lake, Tonle Sap, the size of which is extremely variable depending on the season.

The ring has been breached in several places, and no trace of relief remains at these sites. The surviving portion of the ring, about 60% to 70%, appears to rise a few hundred meters from the surrounding flat floor. The ring hills are quite rugged, yet they lack the characteristic contrast of the local karst formations. The hills appear to be the same color as the floor material. Two characteristics seemed worth noting at the time of observations. The width of the hills is only a small fraction (5 to 10%) of the diameter of the ring; and the demarcation between the ring wall and floor is sharp, almost perpendicular, with little or no grade to the floor as it approaches the ring. The floor inside the ring appears to be a continuation of a once-larger lake bed which extended to the southwest. Looking at the entire region, one has the definite impression that in the past the lake grew and receded, flooding the area of the ring many times. This would seem to explain the sharp boundary between the wall and the floor.

The feature was observed many times; however, this report is the first written record which has been made. No more extensive documentation of the feature was made because the observer believed that such a feature, so prominent from the air, must have been known and thoroughly investigated.

GEOMORPHOLOGY

An enlarged portion of a LANDSAT photo is shown in Figure 1. The possible crater, which may be the source of Australasian tektites, is represented by an incomplete oval ring of hills. The hills rise to a maximum height above the surrounding countryside of about 400 meters (1300 ft.) on the southwest side of the ring. No hills are observed in LANDSAT imagery over most of the northeast quadrant of the ring. The long axis of the oval is 10 km (6.2 miles) long; the short axis is 6 km (3.7 miles). The azimuth of the long axis is about 150° east of north. An airborne observer viewing a 6 X 10 km oval at an oblique angle may well have described it as being remarkably circular, especially if the viewing direction was along an axis of the oval.

A topographic profile through the range of hills on the southwest side of the oval shows generally that from the top of the hills to the surrounding

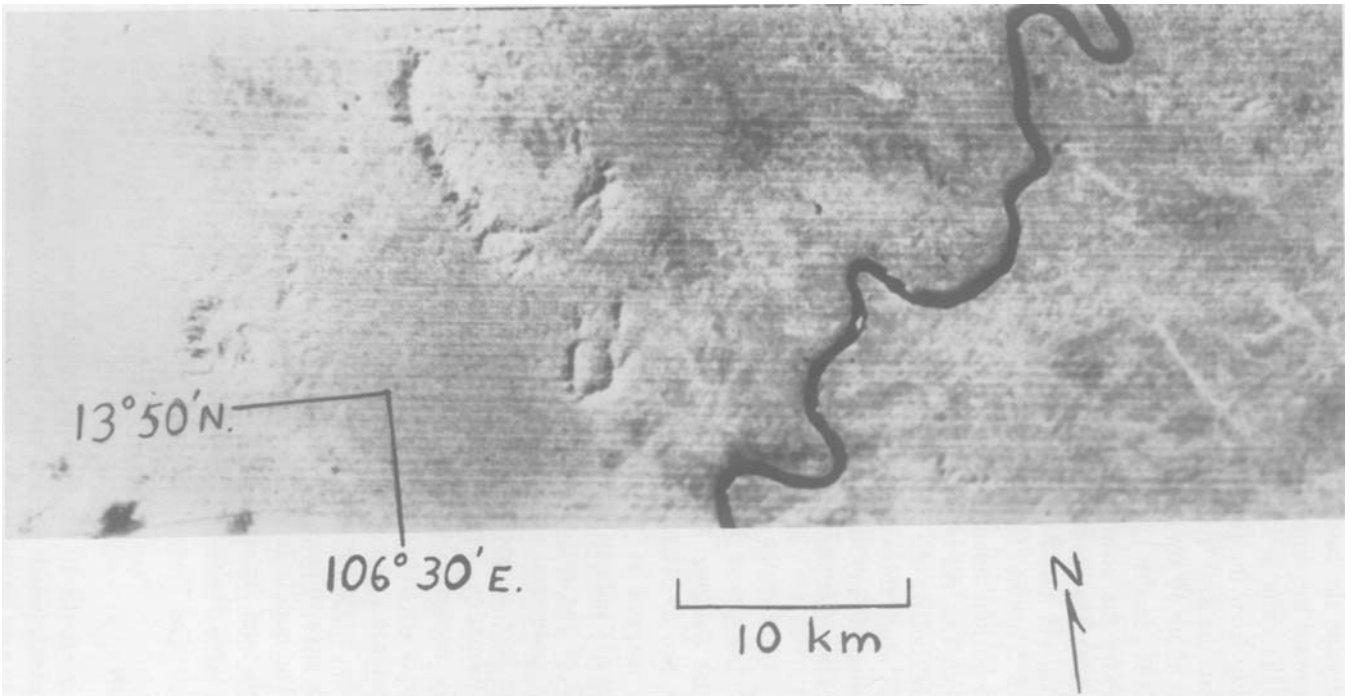


Fig. 1 A portion of LANDSAT infrared photo (ID No. E-1163-02434-7 01) showing a small area in northeastern Cambodia. The structure identified as the possible source of Australasian tektites is represented by the partial oval ring of hills at the lower center of the photo. The ring is about 10 km long and 6 km wide and is located at 106°34'E. longitude and 13°55'N. latitude. North is toward the top of the photo.

level countryside the land slopes more steeply away from the center of the ring than toward the center. The altitude of the land surface inside the ring is essentially the same as that outside the ring. Water drainage is essentially radially away from the ring. The northeast side of the ring is breached to accommodate drainage from within the ring.

Lying about 10 km (6.2 miles) south-southeast of the center of the oval ring of hills is a smaller U-shaped topographic feature, which rises 250 meters (800 ft.) above the countryside. The arms of the U are separated about 1.5 km (1 mile); the azimuth of the axis of the U is about 15° east of north. About 16 km (10 miles) southwest of the center of the oval ring is a single hill, about 370 meters (1200 ft.) high and 6 km (3.7 miles) across at its base. There is no obvious connection between the oval ring, the U-shaped feature, and the single hill. All of these features lie in a relatively level part of the basin which contains the lake, Tonle Sap, about 200 km to the southwest, and through which flows the Mekong river. The present elevation of the lake surface is about 10 m (30 ft.) above sea level and that of the floor inside the oval ring is somewhat less than 150 m (500 ft.).

GEOLOGY

On the *Geologic Map of Asia and the Far East* (United Nations Publication No. 69-30632, 1971) all three topographic features described above are mapped as "Carboniferous – Triassic dacite, rhyolite, and microgranite of Cambodia, Laos, and Viet Nam."

Much of the surrounding countryside consists of relatively recent Quaternary continental sedimentary deposits, thus supporting the idea of a much larger lake at one time. Underlying these deposits are rocks of the Indosinias formation, which also outcrop in the vicinity of the oval ring. These rocks, according to the Explanatory Note for the above-mentioned map, "range in age from middle-late Carboniferous (late Paleozoic) to middle Triassic (Mesozoic) and are predominantly sandstones closely interbedded with massive limestones. In many places, the lower Indosinias is intercalated with large volumes of andesitic, dacitic, and rhyolitic lavas, tuffs, and breccias." The middle Indosinias consists of marine sediments; "sandstones, shales, marls, and limestones," overlain by "continental deposits . . . of sandstones, sandy shales, red shales, and marls, with some conglomerates and breccias."

DISCUSSION

We suggest the 10 km by 6 km oval ring may represent the cratering event, or events, which formed and ejected all Australasian tektites. The available data may be evaluated to obtain support or to reject this hypothesis.

Tektite ages and crater sizes

Australasian tektites are about 0.7 m.y. old. The 15-m.y.-old Moldavites have been associated with the Ries crater, which is 24 km in diameter. The 1.1-m.y.-old Ivory Coast tektites are presumed to have come from the site of the Lake Bosumtwi crater, which is 11 km in diameter. The oval ring in Cambodia covers roughly one-half the area of Lake Bosumtwi, yet the Australasian tektite strewnfield is apparently far more extensive than the Ivory Coast strewnfield. The significantly greater age of Moldavites may account for their relative sparsity despite the larger size of the Ries crater. But Ivory Coast and Australasian tektites are roughly the same age, so this single comparison would indicate a crater larger than 10 km in diameter should exist to mark the source of the Australasian strewnfield. For our hypothesis to remain viable, it is required that factors other than the size of the crater are important in the formation and distribution of tektites. Such factors might be the compressibility of the target material or the approach angle of the impacting object. More compressible target materials may lead to the production of a larger volume of melt during the impact process. An impacting object arriving at a low angle to the target surface may cause ejection of a greater proportion of melted material, as opposed to burying it beneath the crater.

Geochemistry of tektites and crater rocks

The chemistry of tektites has been studied extensively. Taylor (1969) concluded that tektites originated on the earth and were derived from a sediment of subgraywacke composition, the source of which was material of late Paleozoic or Mesozoic age. The chemistry of rocks forming the oval ring and the surrounding material, to our knowledge, has not been studied. Nevertheless, the presence of a craterform structure in a drainage basin containing Quaternary continental sediments and underlain by sandstones, shales, and marls is consistent with a sub-graywacke composition for tektites. In addition, the ages of the sediments and the surrounding and underlying igneous rocks are also consistent with the age requirements for the source of Australasian tektites outlined by Taylor (1969). Unfortunately, the precise location of the breccias in the Indosinias formation mentioned earlier could not be established. Our hypothesis would be strengthened if these breccias are found to occur in association with the oval ring of hills that we have identified.

Crater morphology

Most impact craters on planetary surfaces are remarkably circular. The non-circularity of the structure that we have identified may be taken as

evidence against an impact origin for the structure. However, the possibility that the oval ring of hills may be due to an impact may be salvaged in one of three ways. First, the structure could be a doublet or triplet; that is, it could have been formed by multiple impacts occurring almost simultaneously, thus producing a non-circular crater similar to that found at the Henbury crater field in Australia. Secondly, the asymmetric development of a growing crater may be caused by preferred directions of weakness related to joint directions in the target rocks. This explanation has been given for the squared-off appearance of the Barringer crater in Arizona (Shoemaker and Kieffer, 1974). Finally, an oval crater may be formed by an impact which occurs at a very low angle to the target surface. See on the Moon, for example, the crater Messier, which is about 12 km long and 7 km wide, and which has prominent rays extending 120 km in the direction of the long axis of the crater (Kosofsky and El-Baz, 1970, p. 98-99). Another highly non-circular structure on the Moon, probably formed by impact, is the 200-km-long by 80-km-wide crater, Schiller (Kosofsky and El-Baz, 1970, p. 18).

Another possible problem is the shape of the supposed rim of the crater. In cross-section, the rim of a recently-formed impact crater essentially always slopes more steeply toward the center of the crater. Based on the topographic evidence available, this is not the case for the oval ring of hills. To convert a topographic profile expected for the rim of an impact crater to the observed profile would require the enlarged lake, postulated earlier, to have severely eroded the exterior approaches of the crater rim. The missing quadrant of hills and the absence of relatively easily recognizable ejecta deposits may also be explained as the result of extensive recent erosion and deposition in that area.

Crater location

Perhaps the most favorable aspect of the proposed source of Australasian tektites is its location. O'Keefe (1976) identifies four types of Australasian tektites: 1) microtektites, 2) flanged buttons, 3) splash-form, and 4) Muong Nong. The order in which these types are listed is, first, according to their characteristic sizes; microtektites are smallest, Muong Nong-type tektites are largest. In addition, the ranking reflects a decreasing input of specific energy to the tektite material to achieve the level of melting displayed by the tektites. We suppose that, per unit mass, more of the impact energy was absorbed by microtektites than by flanged buttons, and so on. We also assume a correspondence between thermal energy absorbed and the mechanical energy; that is, ejecta velocity acquired. Thus, the above ranking also reflects a decreasing distance between the location of the tektite type and the source of the tektites. Therefore, we expect microtektites and flanged buttons to be found farthest from the source crater and Muong Nong-type tektites to be found nearest the crater.

According to Barnes (1963) the range of locations where Muong Nong-type tektites have been found extends in a northeast-southwest direction from Muong Nong, Laos, to Kampong Speu, Cambodia, and in a northwest-southeast direction from west of Khon Kaen, Thailand, to Dalat, Viet Nam. These locations may be enclosed by an oval 830 km (520 miles) long and 480 km (300 miles) wide. The long axis of the oval extends in the northwest-southeast direction. A point corresponding to the "center-of-gravity" of these four locations lies 120 km (75 miles) to the west of our proposed crater. Considering the uncertainties in the sampling of Muong Nong-type tektites, we conclude that the proposed crater is located essentially at the center of the Muong Nong tektite strewnfield. For this reason alone, the oval ring of hills in Cambodia should be considered seriously as a candidate for the source of Australasian tektites.

CONCLUDING REMARKS

We have identified an oval craterform structure in southeast Asia which may be the source of Australasian tektites. The size and shape of the structure seem not to be consistent with what we would have expected initially. The geological environment, possible post-impact modifications of the crater, and especially the location of the structure are factors more supportive of our hypothesis. While our arguments are not conclusive, two facts remain: 1) The source of Australasian tektites has not been identified – yet, and 2) Our understanding of geological processes occurring after large impacts on a planetary surface, especially the Earth's, is at best severely limited. Clearly, more data based on analysis of rock samples from the proposed impact site are needed. We would welcome suggestions and assistance leading to acquisition of such samples.

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